



Case study of vortex-induced vibration and mitigation mechanism for a long-span suspension bridge

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ABSTRACT

The wind-induced vibration of long-span bridges mainly manifests as flutter, galloping, vortex-induced vibrations (VIVs), buffeting, etc. Among these, VIVs cause vertical or torsional single-mode vibrations of the main girder; these affect the comfort of drivers or pedestrians on the bridge when vortex-induced resonance occurs. In this study, field measurements, numerical simulations, and wind tunnel tests were used to systematically investigate triggering mechanism during the sudden first and secondary VIV phenomena of Humen Bridge, along with the structural mode and damping characteristics and VIV aerodynamic mitigation effects, etc. The results show that the sudden first VIV was caused by the temporary installation of water-filled barriers, as these changed the aerodynamic configuration of the box girder; the decisive removal of these water-filled barriers can effectively suppress the VIV amplitude to some extent. The succedent secondary VIV was caused by the reduction of the structural damping; therefore, aerodynamic control and damping control measures are proposed. The aerodynamic control measures include the implementation of suppressor plates on the top of handrails and removal of the maintenance rails under the girder, whereas the structural damping control measures involve the successive installation of tuned-mass damper (TMD) to compensate for the reduced damping owing to the former large-amplitude VIV performance. After verification based on reduced-scale model wind tunnel tests, the aerodynamic control measures and structural damping control measures are successfully adopted and applied to Humen Bridge. These measures are shown to effectively suppress VIVs, as confirmed via succedent long-term on-the-spot observations.

1. Introduction

Investigations of the wind-induced vibrations of long-span bridges originated from the Old Tacoma Narrows Bridge in Washington State, USA. Its flutter instability and collapse on November 7, 1940 shocked the world and initiated a brand-new wind engineering discipline, with a specific focus on the wind resistance of long-span bridges. Research over the past 80 years has found that the wind-induced vibration of long-span bridges mainly manifests as four forms: flutter, galloping, vortex-induced vibrations (VIVs), and random buffeting. Among them, flutter and VIVs are the two essential forms of wind-induced instability vibrations, and they are also the common forms lead to the major wind-induced effects threatening the safety and running states of long-span bridges. Flutter instability occurs in two forms - coupled torsion/vertical flutter or pure torsion one (Farquharson et al., 1949; Billah and Scanlan, 1991), and can lead to structural collapse after amplitude

divergence. VIVs cause bending and/or torsional single-mode vibrations of the girders, thereby affecting driver and pedestrian comfort on the bridges during vortex-induced resonance (Simiu and Scanlan, 1996; Xiang, 2005).

Whereas the flutter phenomenon was reported during a wind-related damage accident after the completion of construction of the Old Tacoma Narrows Bridge, VIVs had already appeared during the construction phase of the bridge and continued until the completion of the bridge, and even lasted until a few hours before the destructive flutter occurrence. According to documentary records (Ammann et al., 1941; Scanlan, 1982), before the Old Tacoma Narrows Bridge was destroyed by the flutter, under the actions of strong winds of 16–18 m/s, a fifth-order symmetric vertical mode VIV occurred. The vibration frequency was 0.633 Hz, the single-peak amplitude exceeded 0.40 m, and the vibration lasted more than 2 h. When the wind speed reached 19 m/s, the vertical mode VIV suddenly transformed into a torsional-coupled flutter, which

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was then dominated by the torsional mode. The torsion amplitude rapidly increased to more than 30°. Owing to the central buckling and large-amplitude vibrations of the suspension bridge, the quarter-point suspenders broke first. Subsequently, the suspenders broke continuously along the bridge, ultimately leading to the collapse of the bridge girder.

The locked-in VIVs of long-span bridges usually do not cause structural damage or even collapse; nevertheless, VIV-related events of bridges are frequently reported worldwide. The 300-m span steel box continuous beam bridge built in 1974 in Rio de Janeiro, Brazil suffered an early VIV accident (Battista and Pfeil, 2000). Subsequent continuous beam bridge VIV accidents include those at the Tokyo Bay Bridge in Japan (Fujino and Yoshida, 2002), approach bridge of the Danish Great Belt East Bridge (Schewe and Larsen, 1998), Volga River Bridge in Moscow, Russia (Corriols and Morgenthal, 2012), and Chongqi Bridge in China. These steel continuous beam bridges mainly adopted tuned-mass dampers (TMDs) to mitigate VIV (Xu et al., 2019; Dai et al., 2020). VIV control and mechanism investigations have been conducted on suspension bridges, including the Deer Isle Bridge in the United States (Kumarasena et al., 1991), the Danish Great Belt East Bridge (Larsen et al., 2000), the Xihoumen Bridge in China (Ge, 2011), and the Yi Sun-Shin Bridge in South Korea (Hwang et al., 2020). Among them, the Danish Great Belt East Bridge used a deflector to eliminate vortex shedding at the bottom of the beam and achieve aerodynamic control. The Xihoumen Bridge adopted folding wind barrier to eliminate VIVs, and the VIV of the Yi Sun-sin Bridge in South Korea was stopped after a temporary screen covering the guardrail for construction was removed (Hwang et al., 2020).

With increases in bridge span, the structural rigidity inevitably decreases, the mass becomes lighter, and the damping decreases, resulting in the occasional occurrence of VIVs in long-span bridges, and posing new challenges for bridge wind resistance research (Xiang and Ge, 2011; Li et al., 2011; Zhao et al., 2022). Studies on the VIV issues of main girders generally adopt reduced-scale model wind tunnel tests to predict the actual bridge vortex-related performance (Yang et al., 2021). Owing to the Reynolds number difference (Noguchi et al., 2020), structural damping ratio (Hwang et al., 2020), and oncoming turbulence (Zhu et al., 2018), the prediction error for the VIV amplitude is relatively large; moreover, the VIV amplitude is the main indicator for deciding whether to adopt VIV control measures. The VIV resonance values of the scaled model and actual bridge not only depend on external loading factors such as the wake vortex mode or vortex shedding frequency (Hu et al., 2021), but also have a close relationship with internal structural factors, such as the mode damping ratios of real bridge (Chu et al., 2021). Although the aerodynamic configuration can be optimized in advance by weakening the influences of external factors, the internal factor of a real bridge damping ratio can only be identified after the completion of construction, and there are considerable damping measurement errors and identification uncertainties. In fact, long-span bridges inevitably suffer from wind-related stochastic vibrations or VIVs (Zhu et al., 2018); the only difference is in the size of the amplitude. Up to now, it is rather hard to provide a convincing safety limit standard for the VIVs of long-span bridges (Ge et al., 2019).

A series of methods such as theoretical analyses (Xu et al., 2017), field measurements (Li et al., 2014), wind tunnel tests (Yang et al., 2021) and numerical simulations (Noguchi et al., 2020) have been used to reveal the causes of VIVs, develop aerodynamic or damping control measures, and describe the principles of aerodynamics and damping control. Nevertheless, it was precisely under this background of VIV research on long-span bridges (Xu et al., 2017; Hu et al., 2018; Hu et al., 2019), after 23 years of normal operation of the Humen Bridge Suspension Bridge, first and secondary VIVs suddenly occurred, involving three types of vertical modes. To address this thorny problem, a fast on-site measure-analyze-suppression response to control such VIVs was conducted (Zhao et al., 2022), and quick and easy preliminary aerodynamic measures and damping measures were creatively adopted to

mitigate the VIV response and resume traffic as soon as possible. Long-term aerodynamic control measures and damping control measures for completely eliminating the VIV resonance have also been suggested and optimized.

2. Dynamic characteristics and vortex-induced vibration performance

The Humen Bridge is a single-span suspension bridge with a main span of 888 m. The bridge tower is a portal-type reinforced concrete structure. Both ends of the main cable adopt gravity anchors. The mid-span rise-span ratio is 1/10.5, and the main girder adopts a streamlined closed steel box type with a width of 35.6 m and height of 3.0 m. Vertical bearings and lateral wind-resistant bearings are set at the two pylons. The main cable and the stiffening girder are connected by steel strand hangers; the hanger spacing is 12.0 m, and the full bridge has 72 pairs of hangers, as shown in Fig. 1 (Niu, 1998).

Construction of the Humen Bridge began on October 28, 1992, and the bridge was opened to traffic on June 9, 1997. It has been in operation for nearly 23 years. Before the suspension bridge was completed and opened, a comprehensive completion acceptance test was conducted. Combined with a wind resistance research report from the design stage, calculation and test results were obtained for the structural dynamic characteristics of the first five vertical modes and first two torsional modes, as shown in Table 1 (Zhang and Zhu, 1999). The modeling and calculation of the dynamic characteristics was based on the "SAP5" commercial software. With the help of acceleration sensors, the dynamic characteristics were identified using the half-power spectrum method under random excitation. The measured damping ratios of the second-order torsional mode and fifth-order vertical mode were missing, owing to certain technical reasons.

In general, a sectional model wind tunnel test needs to simulate the mass-related parameters, in which the main cable, hangers, and bridge tower are converted to be equivalent to the mass and mass moment of inertia of the main girder. Therefore, according to the principle of physical similarity for structural wind resistance design, the equivalent mass or equivalent mass moment of inertia are calculated as follows:

$$m_{eq,i} = \frac{\int m \phi_i(x)^2 dx}{\int_D \phi_{hi}(x)^2 dx} \quad (1)$$

$$I_{eq,i} = \frac{\int I \phi_i(x)^2 dx}{\int_D \phi_{ai}(x)^2 dx} \quad (2)$$

In Eqs. (1) and (2), $m_{eq,i}$ and $I_{eq,i}$ represent the equivalent mass and mass moment of inertia of the i th order mode shape, respectively; m and I are the mass of the structural member and mass moment of inertia, respectively; i represents the vibration mode number of the structural member; and ϕ_{hi} and ϕ_{ai} represent the i th order vertical mode and i th order torsional one of the main girder, respectively.

Since the locked-in wind speed ranges and amplitudes of the VIVs of a bridge are closely related to the Scruton number (S_c), the S_c number is generally used to evaluate the VIVs of a bridge. The S_c number is proportional to the structural mass m and damping ratio ξ and inversely proportional to the square of the bridge deck width B , as follows:

$$S_c = \frac{4\pi\xi m}{\rho B^2} \quad (3)$$

The larger the value of S_c , the less likely VIVs occur. In contrast, vortex-related resonance is more likely. The S_c numbers of the first seven orders are listed in Table 1. Among them, the damping ratios of the first five-order modes are the average measured values. The damping ratios of the last two order modes are absent due to technical reasons, assumed as 0.50% referred to proposed value from Wind-resistant Design Specification for Wind Highway Bridges (2004). The second-order torsional mode uses the mass moment of inertia I divided by the square of the

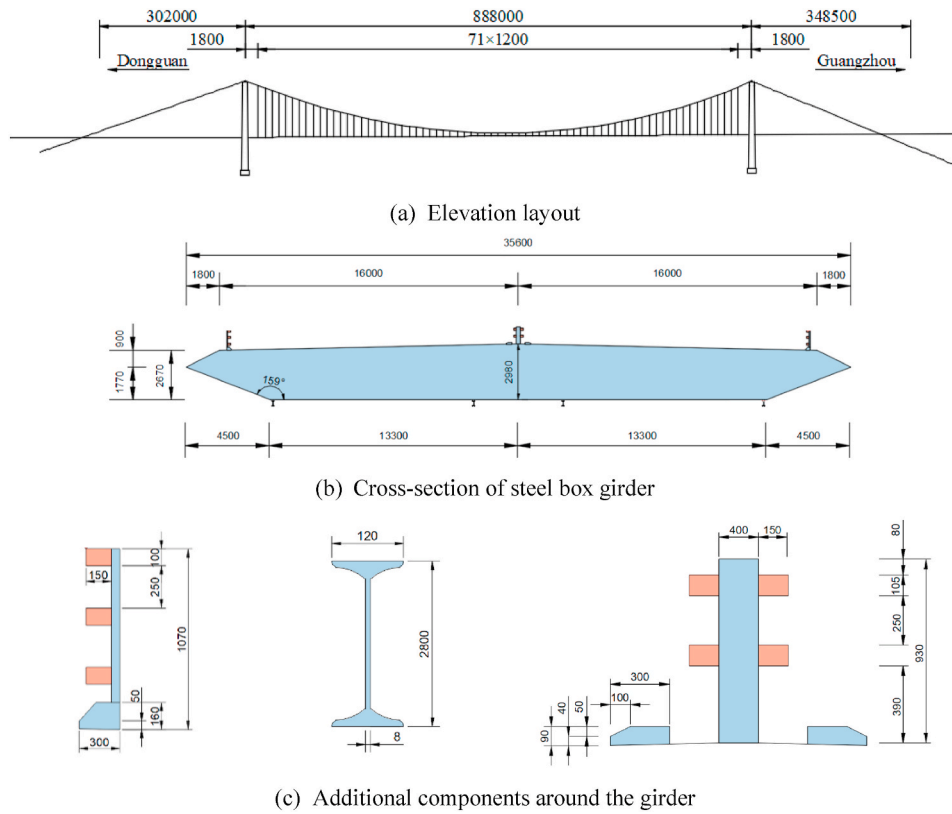


Fig. 1. Overall layout of the Humen Bridge (Unit: mm).

Table 1
Main structural dynamic characteristics of the Humen Bridge identified at completion stage.

No.	Natural vibration frequency/Hz		Main beam mode Description	Damping ratio measured/%		Equivalent mass or mass moment of inertia/(t/m or t-m ² /m)	S_c number/($4\pi\xi m/\rho B^2$)
	Calculated value	Measured value		Range	Mean value		
1	0.1117	0.1344	First-order vertical	0.79–2.48	1.32	43.72	4.58 > 1.0
2	0.1715	0.1705	Second-order vertical	0.56–1.37	0.88	23.13	1.61 > 1.0
3	0.2251	0.2325	Third-order vertical	0.50–1.20	0.70	25.03	1.39 > 1.0
4	0.2765	0.2768	Fourth-order vertical	0.49–1.02	0.63	24.38	1.22 > 1.0
5	0.3306	0.3321	First-order torsional vibration	0.20–0.65	0.50	5552	2.33 > 1.0
6	0.3612	0.3748	Second-order torsional vibration	–	–	3518	1.47 > 1.0
7	0.3682	0.3687	Fifth-order vertical	–	–	25.04	0.99 < 1.0

radius of inertia r instead of the mass m , i.e., I/r^2 . It is not difficult to find that, except for the last order mode (the fifth-order vertical mode), the S_c values calculated according to the average value of the measured damping ratio are all greater than 1.0, proving that according to the structural damping ratio at the completion stage, the suspension bridge is less prone to VIV risks, except for the fifth-order vertical mode.

3. Field measurement of VIVs and triggering mechanism

The Humen Bridge suffered a sudden vertical vibration of the bridge deck in the afternoon on May 5, 2020. The vibration continued after the bridge deck was closed to traffic, and was calmed down after the removal of temporary water-filled barriers which has been added for the hanger-maintenance construction.

However, in the following hours, secondary vertical vibrations were perceivable on the bridge girder. These were determined by field measurement and theoretical analysis as high-order VIVs with 3 vertical modes, involving fifth- fourth- and third-order vertical mode resonance,

and were mainly dependent on the wind conditions of the bridge location and decreasing structural damping ratio, as shown in Table 2.

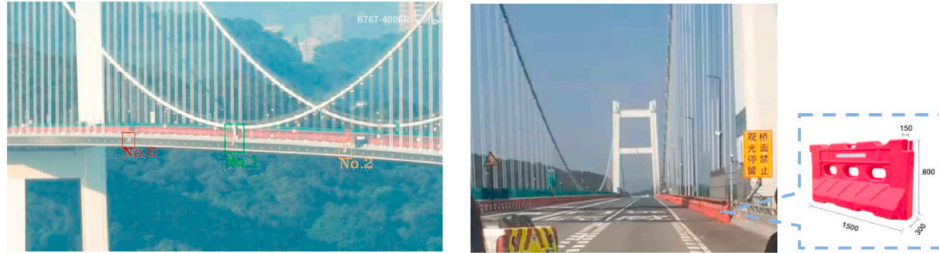
3.1. Fifth-order vertical VIV

While the VIVs began at first on the afternoon of May 5, there are not any monitoring system or sensors on the bridge though the bridge has run for 23 years. The recorded video and digital inversion technology were utilized to identify time-variant displacement responses and phase lags for 3 key points (see Fig. 2(a) and (c)), in which No.0-No.2 are approximately 3/8, 1/2 and 5/8 span, respectively. Then the main girder of the Humen bridge were judged to undergo vertical vortex-induced resonance on May 5, and the vibration mode was the fifth-order vertical one with the frequency of 0.367 Hz. Although the VIVs lasted for approximately 6 h from 12:30 p.m., owing to the influences of the wind speed and wind direction, the VIV amplitudes were occasionally large and small, showing a time-variant clustered wave type. The VIV mode and acceleration time history records from the acceleration

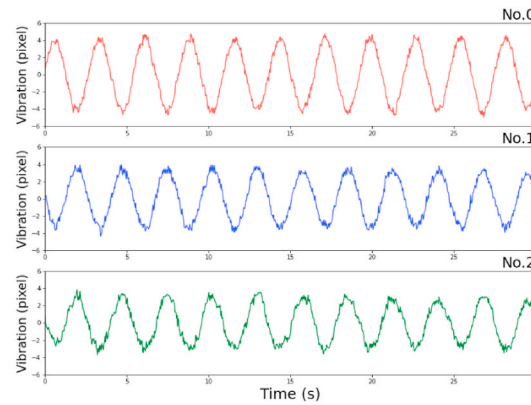
Table 2

Field measurement of vortex-induced vibration characteristics of the Humen Bridge.

No.	Frequency measured/Hz		Main girder mode	Damping ratio measured/%		S_c number	Locked in Wind speed/(m/s)	Acc./(m/s^2)	Disp./m
	Upon completion	After VIVs		Upon completion	After VIVs				
1	0.369	0.367	Fifth-order vertical	0.50	0.25	0.51	10.7–12.8	1.30	0.310
2	0.277	0.276	Fourth-order vertical	0.63	0.28	0.54	8.3–9.6	0.84	0.278
3	0.233	0.230	Third-order vertical	0.70	0.26	0.52	6.3–8.5	0.48	0.230



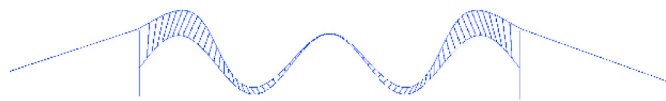
(a) Displacement monitoring points along the girder (b) Installation of water-filled barriers



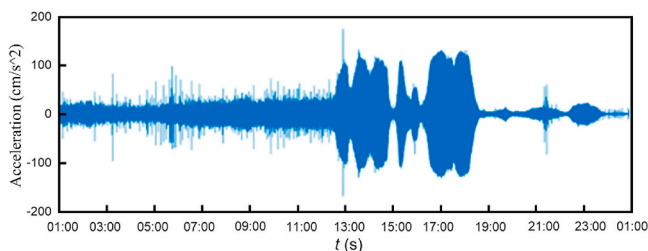
(c) Vertical vortex-induced vibration response

Fig. 2. Sudden VIVs of the Humen Bridge during temporary construction.

sensors on the bridge are shown in Fig. 3. The maximum vertical acceleration as measured at the mid-span of the main girder was 1.30 m/s^2 , and the corresponding vertical displacement was 0.310 m .



(a) Fifth-order vertical vibration mode



(b) Time history of acceleration in mid-span of main beam

Fig. 3. VIV record of the fifth-order vertical mode.

To explain the mechanisms of the VIVs of the suspension bridge, it was first confirmed that when the vibrations occurred, the measured wind direction on the bridge deck was a southeast wind. The oncoming wind direction was almost perpendicular to the main girder with declination angle of $90^\circ \pm 7^\circ$. While the VIVs occurred, the measured wind speed on the bridge deck was $10.7\text{--}12.8 \text{ m/s}$, with an average value of 11.8 m/s . In additional, the bridge deck was undergoing a hanger repair construction. The temporary water-filled barriers formed a continuous wall for an approximately rectangular section with a height of 0.8 m and width of $0.15\text{--}0.3 \text{ m}$. The entire temporary barrier extended continuously from the one-fourth span to the tower for more than 600 m , seriously damaging the aerodynamic configuration of the streamlined box girder.

The flow field information around the bridge girder with or without water-filled barriers would be very helpful to clarify the VIV-related mechanism. Up to now, it is usually very difficult to accurately measure the fluid structures of a flow field whatever in wind tunnels or under the on-situ, because it usually requires very expensive instruments, e.g., PIV (Particle Image Velocimetry). To obtain characteristics of flow structures when a bridge girder is undergoing VIV, we employed Computational Fluid Dynamics (CFD) method to re-illustrate the flow field state, in which Large eddy simulation (LES) using the Smagorinsky subgrid turbulent model was employed to solve the flow field around the bridge girders. CFD numerical simulations of the flow

development around the girder were conducted at Reynolds number of about 8.0×10^4 , corresponding to wind speeds of 8 m/s. Every mesh was refined to meet mesh convergence, and the initial value of y^+ was taken as 1.72, and the results comprised a total of 145,191 mesh elements. A triangular mesh element was adopted in the numerical simulation analysis, and an unstructured mesh was used herein for computation domain discretization to considering the complexity of the bridge geometry and the need for an efficient simulation resolution time.

Fig. 4 shows the boundary conditions and computational domain setting. Fig. 5 illustrates the mesh scheme and computation domain discretization, with a height of 104.8 m and width of 226.4 m, which provides a blockage ratio of 2.8%. The length of the oncoming flow zone is 20 times the height of the bridge girder, the length of the wake region is 50 times, and the upper and bottom boundaries are all 17 times. In addition, flow around the bridge deck is considered as incompressible uniform flow and a stationary model utilized in CFD simulations. The entrance condition of the computation domain was set as a velocity boundary condition and the free outlet boundary was taken as the outlet boundary condition. The non-slip solid wall boundary condition was taken as the boundary condition of the bridge section. And the $P0$ and $U0$ are the initial wind speed and pressure of the inlet boundary conditions. The SIMPLE pressure-velocity coupled method was adopted to solve the pressure-velocity coupled field, which had mature application in an atmospheric boundary layer (Argyropoulos and Markatos, 2015).

Numerical analysis of the fluid dynamics indicated that double large energy vortices were formed on the windward side along the above- and below-bridge deck, thereby exciting the vortex-induced resonance, and that the temporary water-filled barriers strengthened the vortex intensity above the girder, as shown in Figs. 6 and 7. The fifth-order vertical mode damping ratio decreased from 0.50% (as assumed in Tables 2) to 0.25%, and the Sc number calculated on this basis dropped from 0.99 to 0.51. The lower damping ratio makes VIVs possible even under a lower-energy vortex excitation though removing the temporary water-filled barriers, especially the separated vortex generated around upstream maintenance rail. These factors precisely contribute that the fifth-order vertical VIV suddenly occurred on the afternoon of May 5, even the water-filled barriers were removed thoroughly.

3.2. Fourth-order vertical VIV

After the Humen Bridge suddenly suffered from the fifth-order vertical VIV on the afternoon of May 5, the large-amplitude vibration was suppressed while removing the temporary water-filled barriers on the lateral side of the bridge deck. However, the secondary vortex-induced resonance appeared again, mainly manifesting as the fourth-order vertical mode, and the measured vibration frequency was 0.276 Hz. Moreover, as long as the wind speed and wind direction conditions were in the ranges of specific values, harmonic VIVs could appear at any time, but the amplitudes varied either large or small. The VIV mode and acceleration records from the newly arranged acceleration sensors on the bridge are shown in Fig. 8. The maximum vertical acceleration and displacement measured at the three-eighth point of the main beam was

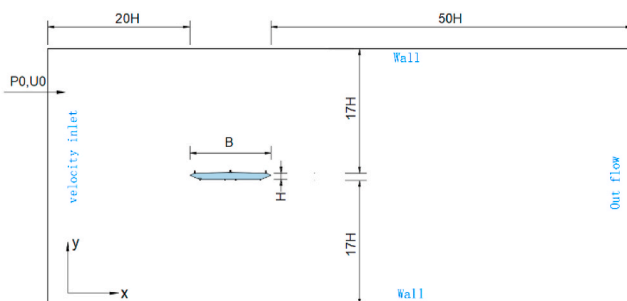
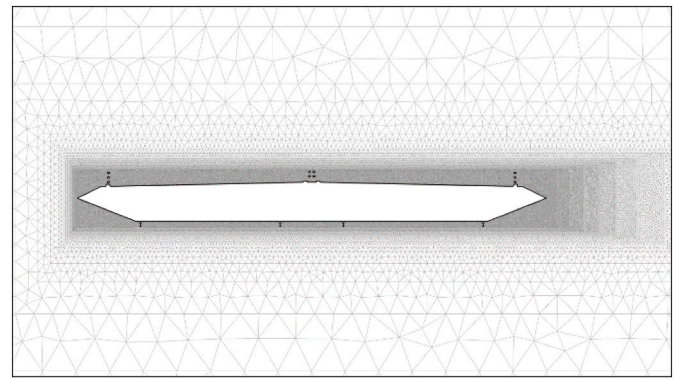
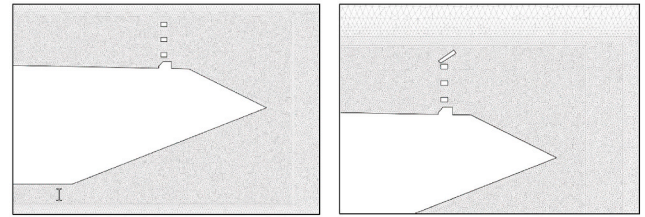


Fig. 4. Boundary conditions and computation domain setting.

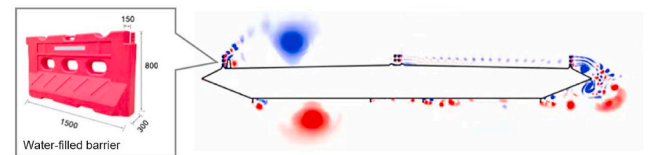


a) Mesh scheme around bridge deck

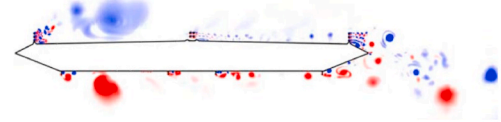


b) Local view of refined mesh scheme

Fig. 5. Mesh scheme and computation domain discretization.



(a) Impact of temporary water-filled barriers



(b) Removal of temporary water-filled barriers

Fig. 6. Numerical analysis of flow around box girder of suspension bridge.

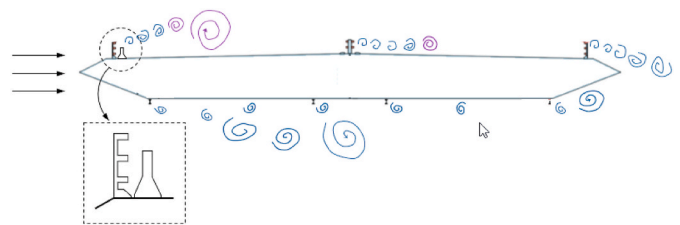
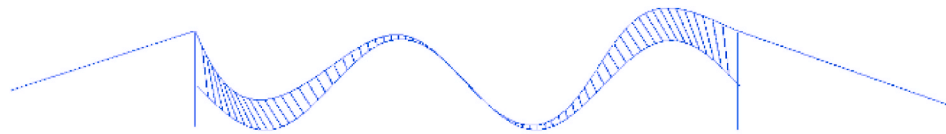


Fig. 7. Triggering mechanism of strengthened vortex-shedding nearby upstream water-fill barrier and maintenance rail.

0.84 m/s^2 and 0.278 m, respectively.

When the VIV occurred in the bridge girder, the measured wind direction of the bridge deck was at 5° from the perpendicular axis of the bridge. The measured wind speed on the bridge deck was 8.3–9.6 m/s with an average value of 9.0 m/s, also meeting the lock-in conditions of VIVs (see Fig. 11 (a)). A CFD numerical analysis showed that the windward side of the original section also had a double vortex with stronger energy (see Fig. 6(b)), acting as an external cause of the vortex-induced resonance. The test results regarding the damping ratio



(a) Fourth-order vertical mode

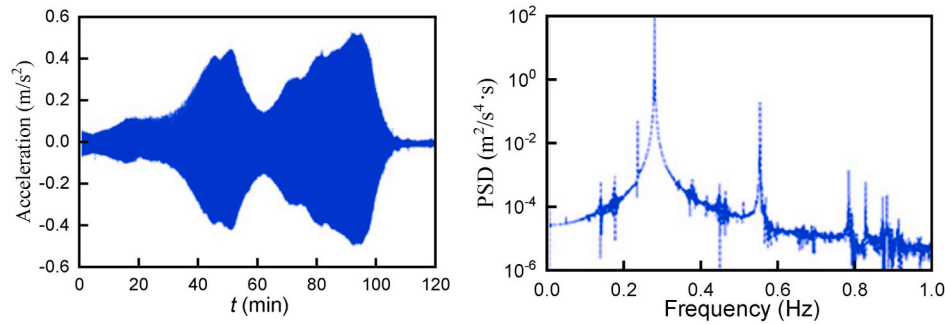
(b) VIV acceleration response at three-eighth point of the main beam
(left: time history; right: power spectrum)

Fig. 8. Fourth-order vertical VIV.

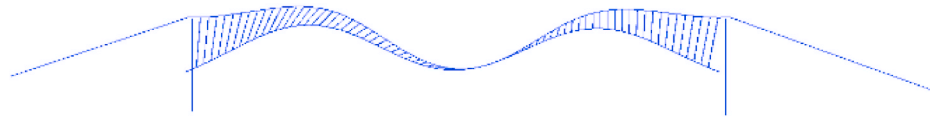
indicated that the fourth-order vertical mode damping ratio dropped from 0.63% to 0.28%, as shown in Table 2. The S_c number calculated on this basis dropped from 1.22 to 0.54. The decrease in the damping ratio was the main cause of the fourth-order vertical VIV.

3.3. Third-order vertical VIV

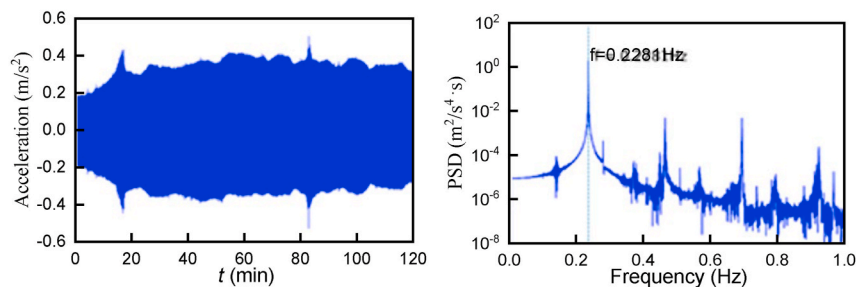
Along with the fourth-order vertical mode VIV, while the wind speed decreased, a third-order vertical mode VIV was observed simultaneously, with 0.230 Hz vibration frequency. Similar to the fourth-order vertical vibration, as long as the wind speed and wind direction conditions were favorable for the VIV locked-in condition, the harmonic vibration could appear at any time, and the amplitude similarly showed a non-stationary state. The VIV mode and acceleration time history

records from the acceleration sensors arranged on the bridge are shown in Fig. 9. The measured maximum vertical acceleration and displacement of the VIV at the quarter point of the main girder was 0.48 m/s^2 and was 0.230 m, respectively.

When the VIV occurred in main girder of the suspension bridge, the measured wind speed on the bridge deck was 6.3–8.5 m/s, with an average value of 7.4 m/s; this was also in line with the wind conditions for VIVs. The damping ratio test of the suspension bridge structure indicated that the third-order vertical mode damping ratio dropped from an average value of 0.70%–0.25% (see Table 2). The S_c number calculated on this basis decreased from 1.39 to 0.52. Similarly, the decrease in the damping ratio was the main cause of the third-order vertical VIV.



(a) Third-order vertical vibration mode



(b) VIV acceleration response at quarter point of main beam

Fig. 9. Third-order vertical VIV.

4. Comparison and verification of VIV response by wind tunnel test

At present, the most reliable method for wind-induced vibration research on long-span bridges is the use of a reduced-scale model wind tunnel test. When studying VIV performance, a large-scale sectional-model wind tunnel test is generally adopted. During the emergency case of the Humen Bridge, several wind tunnel divisions, including those of Tongji University, Southwest Jiaotong University, and Chang'an University, conducted wind tunnel tests with four types of sectional models, covering various scaled ratios of 1:60, 1:50, and 1:20. The VIV response measured on the actual bridge was also reproduced and verified, laying the foundation for aerodynamic control and damping mitigation of the VIVs. In this section, the VIV test conditions only aim at the actual original bridge section without additional aerodynamic measures.

4.1. VIV wind tunnel test of sectional model

A wind tunnel test of a sectional model mainly simulates the vibration characteristics of the two degrees of freedom (DOFs) of the vertical and torsional modes. In this study, a rigid body sectional model with elastic suspension was adopted, as shown in Fig. 10. Free vibration tests on a spring-suspended sectional model (SSSM) were employed to obtain the wind-induced displacements. The SSSM was installed on a frame-supported system in a wind tunnel. End walls were attached to each side to minimize the end effects, and two laser displacement sensors were attached to each end. The outer arms at both ends of the rigid sectional model were suspended on a wind tunnel support with four springs. In general, in addition to satisfying the geometric similarity requirements, the wind tunnel test of a spring-suspension 2-D SSSM should also meet other similarity criteria, including scaling parameters of damping ratio, frequency and wind speed, etc.

The main dimensions of the girder sectional model included the width B and height H , which were determined according to the geometric scale ratios. The actual bridge girder equivalent mass m was selected from the minimum value of the first five vertical modes, and the equivalent masses of the sectional models with different scaling ratios

were determined according to the square of the geometric scaling ratio λ_L . The actual equivalent mass moment of inertia J_m of the bridge girder, and the equivalent mass moment of inertia of the sectional model was calculated according to the fourth power of the geometric scale ratio λ_L . The vibration frequency of the model referred to the fourth-order vertical vibration and second-order torsional vibration frequency of the actual bridge, respectively, and the vertical and torsional vibration frequencies of the sectional model were determined according to a certain frequency scale ratio. In general, the actual bridge average wind speed scaling ratio λ_U should be equal to the product of the geometric scaling ratio λ_L and frequency scaling ratio λ_f ; here, it was selected according to the wind tunnel test conditions of the sectional model, see Table 3. The structural damping ratio of the third-order vertical modes of the actual bridge was approximately 0.26%; in general, the damping ratio of the sectional model system should be kept below the minimum damping ratio as much as possible to ensure the engineering safety. So, lower damping ratios were adopted during the wind tunnel, and other main parameters of the sectional models for the VIV test are shown in Table 3.

The wind tunnel test of the sectional model only considered an equivalent mass and an equivalent mass moment of inertia, and the corresponding vertical and torsional vibration frequencies. Therefore, three different vertical vibration modes of the actual bridge VIVs were simulated based on a frequency scaling ratio and wind speed scaling ratio. Various wind angles of attack (AoAs) were tested for each aerodynamic configuration condition, namely $\pm 5^\circ$, $\pm 3^\circ$, and 0° . To compare the effects of different damping ratios on the VIV performance, parametric sensitivity tests about structural damping ratios were conducted. The following mainly introduces the results from the wind tunnel test of the 1:50 and 1:20 sectional models.

4.2. VIVs for various sectional models

In order to clarify the VIV response for various reduced-scale models, detailed vertical and torsional performance, including vibration amplitude and locked-in wind speed range, was compared at various AoAs. To reproduce the vertical VIVs of the main girder, the VIV wind tunnel test of the 1:50 sectional model was implemented first, in which the value on vertical and horizontal axes have been converted to the full-scale results. Fig. 11 illustrates the results from the vertical and torsional performances of the sectional model. As shown in Fig. 11(a), the locked-in wind speed range of the vertical VIV of the sectional model is 6–12 m/s, and all five AoAs have large-amplitude VIV responses. The maximum amplitude under the 5° AoA is 0.283 m, and the minimum value under -5° AoA is 0.140 m. According to China's Code for Design of Wind Resistance of Highway Bridges (JTG/T 3360-01-2018), the strictest permitted value about the vertical amplitude limit is 0.145 m during the bridge VIV stage. In Fig. 11(b), the locked wind speed range of the torsional VIV of the sectional model is 10–20 m/s, and all wind AoAs show large-amplitude torsional VIVs. The maximum amplitude of the $+3^\circ$ AoA is 0.99° , and the minimum of the 0° AoA is 0.35° . According to China's Code, the limit of the amplitude of a torsional VIV in bridges is 0.35° .

To increase the Reynolds number and reduce the scale effect under the wind tunnel tests, an extended test of the 1:20 sectional model was further used to simulate the VIV performance. Taking the 8.0 m/s wind velocity as a sample, for 1:50 to 1:20 reduced scale models, the Reynolds number lies between 3.23×10^4 and 8.08×10^4 . The Reynolds number in ordinary wind tunnel tests of the bridge deck has been generally 10^2 to 10^3 times smaller than that at actual bridges, and steady aerodynamic coefficients and unsteady aerodynamic coefficients of bridge deck were reported to be remarkably dependent on Reynolds number (Schewe and Larsen, 1998; Matsuda et al., 2001). In spite of those limitations, previous researches indicate that conventional wind tunnel test results in the low Reynolds number region are conservative for wind resistance design of bridge decks (Matsuda et al., 2001; Larsen et al., 2008). Fig. 12

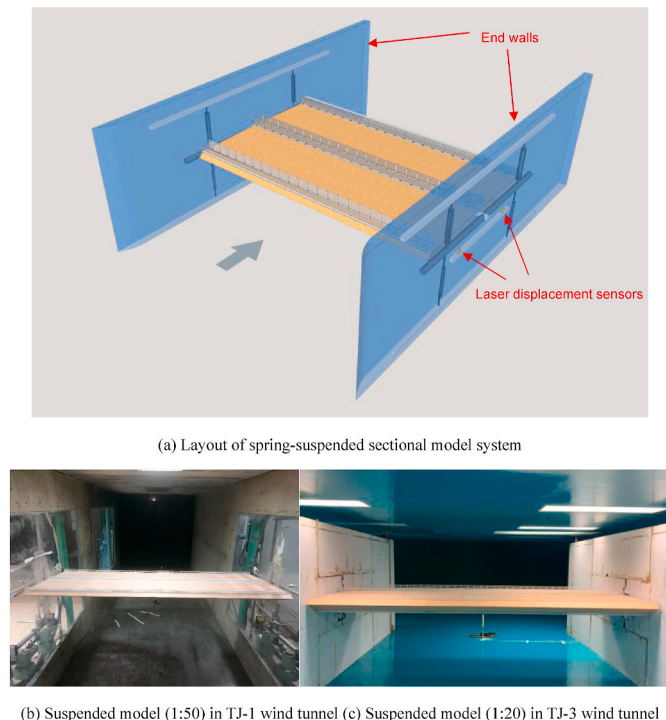


Fig. 10. Schematic diagram of spring-suspended sectional model systems.

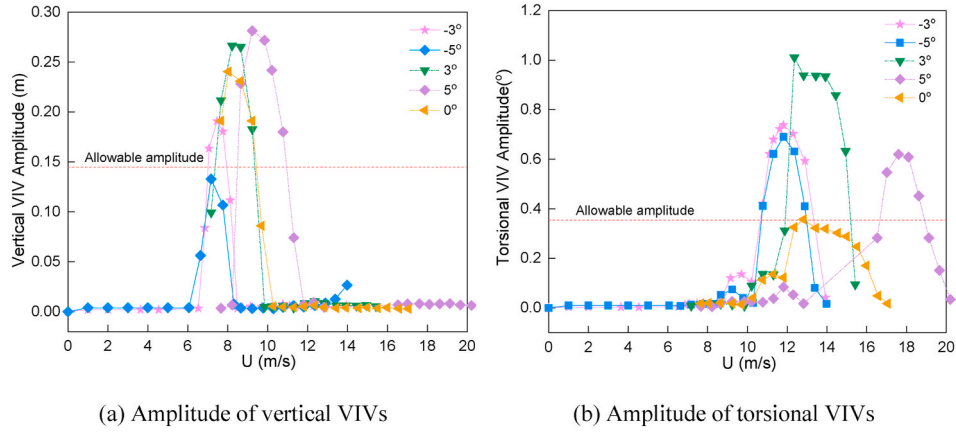


Fig. 11. Wind tunnel test results of 1:50 model.

Table 3

Main parameters of the test sectional model of the Humen Bridge.

Parameter	Variable	Unit	Actual Bridge Value	Reduced scale λ_L		
				1:60	1:50	1:20
Main girder size	Width	m	35.6	0.59	0.712	1.78
	Height	m	3.00	0.05	0.060	0.15
Equivalent mass	Mass	kg/m	23134	6.426	9.253	57.83
	Mass moment of inertia	kg-m ² /m	5551971	0.4310	0.8883	34.70
Vibration frequency	Vertical mode	Hz	0.2765	3.13	4.50	2.78
	Torsional mode	Hz	0.3612	4.09	5.93	4.62
Average wind speed	Wind speed ratio λ_U	–	1:1	1:5.3	1:3	1:2
Minimum damping ratio	Vertical mode ξ_h	%	0.25–0.28	0.22	0.18/0.32	0.15
	Torsional mode ξ_t	%	0.11–0.50	0.29	0.11/0.18	0.15

shows the results from the vertical and torsional VIV responses under the five AoAs. As shown in Fig. 12(a), the locked-in wind speed range of the vertical VIV of the model is 6–10.5 m/s. Vertical VIVs similarly appear in all AoAs, among which the amplitude under the 3° condition is 0.247 m, and the smallest (of –5°) is 0.105 m. The torsional VIV of the sectional model in Fig. 12(b) is not evident, indicating that the torsional mode VIV response is not observed in the actual bridge under the higher Reynolds condition; this can be validated using measured data from full-scale field

observations.

4.3. VIVs between sectional models and full-scale bridge

To convert the wind tunnel test results from the sectional model to the third-order vertical mode of the actual full-scale bridge where the VIVs occurred, the locked-in wind speed and amplitude of the VIVs obtained from the wind tunnel test need to be corrected. The VIV-related correction coefficient η_U can be calculated according to Eq. (4).

$$\eta_U = \frac{\lambda_U}{\lambda_L \lambda_f} \quad (4)$$

Here, λ_U , λ_L , and λ_f represent the wind speed, geometric, and frequency similarity ratios, respectively. They can be calculated according to the parameters listed in Table 3, where the wind speed correction coefficients η_U for the third-, fourth-, and fifth-order vertical modes are 0.827, 0.993, and 1.320, respectively. The VIV amplitude correction mainly involves two aspects: an equivalent mass correction η_m and vertical mode correction η_φ . Correction coefficient calculation formulas can be used, as follows (Zhang et al., 2014):

$$\eta_m = \frac{m_i}{m_o} \quad (5)$$

$$\eta_\varphi = \frac{\varphi_{\max} \int_0^L |\varphi(x)| dx}{\int_0^L \varphi^2(x) dx} \quad (6)$$

Here, m_i and m_o represent the equivalent mass of the i th mode shape and equivalent mass of the sectional model, respectively. The value can be selected according to the equivalent mass in Table 1; $\varphi(x)$ represents

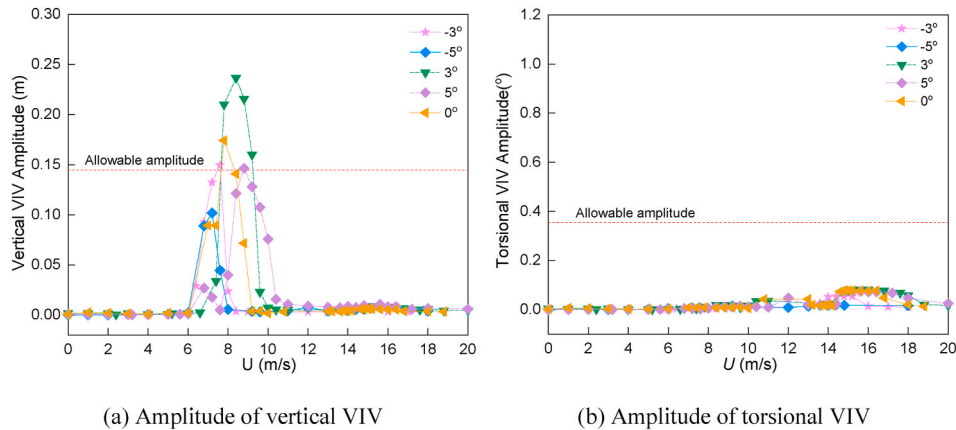


Fig. 12. Wind tunnel test results of 1:20 model.

the vertical mode function, and the mode correction coefficient of the simple harmonic vibration is $\eta_\varphi = 4/\pi = 1.273$.

According to the above VIV locked wind speed and VIV amplitude correction method, the vertical VIV amplitude and locked wind speed obtained from the wind tunnel test of the 1:20 sectional model can be converted and compared with the actual measured results from the vertical vibration response of the real bridge. Table 4 shows the comparison results. Among them, the third-order and fourth-order vertical VIV amplitudes are greater than the actual measured results from the actual bridge. This is mainly because the damping ratio of the sectional model is only 0.15%, whereas the damping ratio of the actual bridge structure is between 0.25% and 0.28%. The fourth-order vertical VIV amplitudes are smaller than those from the actual bridge measured results. This is mainly because the existence of the water-filled barriers increases the VIV excitation. In addition, the model test results are consistent with the actual measured results from the full-scale bridge.

5. VIV aerodynamic control and mechanism

In addition to using structural measures to suppress excessive vibrations, the VIV control methods for long-span bridges also mainly emphasize aerodynamic control approaches, i.e., by means of flow disturbance and optimization. To mitigate the large-amplitude VIVs of the Humen Bridge, two aerodynamic measures were proposed, namely, the addition of flow-suppressing plates and the removal of the outer maintenance rails. To validate the control effects of aerodynamic measures, multiple-scale sectional model tests conducted in this study, i.e., by adding flow-suppressing plates to the lateral barriers above the main girder, removing the outer maintenance rails.

5.1. Flow-suppressing plate

To resume traffic as soon as possible, the vertical VIV amplitude must be suppressed quickly. For the Humen Bridge, an aerodynamic measure was first attempted by referring to the literature (Hu et al., 2018, 2019), that is, by adding a flow-suppressing plate on the top of the guardrail on both sides of the bridge deck, as shown in Fig. 13. The baffle had a width of 0.4 m and was inclined at 45° with the horizontal plane. It was fixed along the guardrails on both sides of the bridge deck with a full span of 888 m. It only took a day and a half to install all of the flow control plates, allowing the Humen Bridge's traffic to resume in the shortest possible time.

The VIV control effect of the flow-suppressing plate was verified using a wind tunnel test. Fig. 14 shows the vertical VIV test results from the 1:50 and 1:20 sectional models with the addition of flow suppressors. The results from the 1:50 model in Fig. 14(a) show that the VIV continues to exist at the $\pm 3^\circ$ and 0° AoAs, and that the vibration locked-in wind speed is 6.5–9.5 m/s. The maximum amplitude of the VIV at -3° and 0° AoAs is 0.175 m and 0.140 m, respectively, and the minimum amplitude at 3° AoA is 0.115 m. Although the VIV amplitude is evidently reduced relative to that before the addition of the suppressor plate, the control effect still needs to be updated. As shown in Fig. 14(b), the VIV

test results for the 1:20 sectional model show that the VIV disappears under the -3° and -5° AoAs, and the locked-in wind speed is 7–10 m/s. The maximum amplitude of the VIV at the 3° AoA is reduced from 0.247 m to 0.156 m, and the amplitude is reduced by 37%. The maximum amplitude at the 5° AoA is reduced from 0.150 to 0.120 m, and the amplitude is 20% smaller. Therefore, the aerodynamic control measures of the suppressor plate can reduce the vibration amplitude by 20%–37%. However, the amplitude of the 0° AoA remains basically unchanged at 0.167 m, showing the maximum amplitude of the VIVs.

5.2. Removal of maintenance rails

To completely suppress the vertical VIVs of the main girder of the suspension bridge, an aerodynamic control measure for eliminating the windward vortex below the girder (see Figs. 6 and 7) was proposed to improve the large-amplitude VIVs, that is, by removing the two outside maintenance rails along the girder bottom without the additional suppressor plates, as shown in Fig. 15. In the real bridge, the outer maintenance rails were set at the edge of the horizontal bottom plate of the box girder. The center was only 0.28 m away from the edge, the height was 0.28 m, the height of the connection with the bottom of the girder was 0.10 m. The demolition of the rails was technically difficult, and required a long construction time. Therefore, it was considered as a long-term measure to proceed after the Humen Bridge resumed traffic.

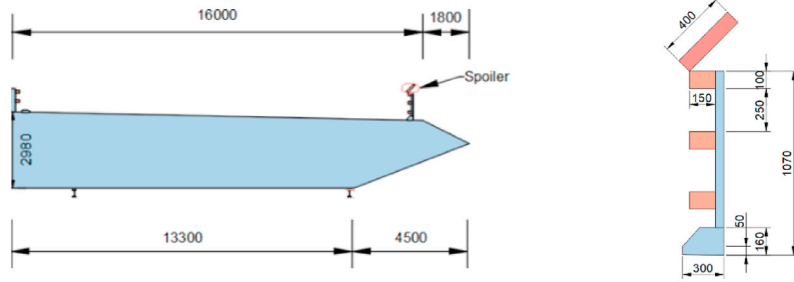
In this study, the aerodynamic control effect of the dismantled maintenance rails was tested based on a wind tunnel test. Fig. 16 shows the vertical VIV results for the 1:50 and 1:20 sectional models after removing the maintenance rail. The test results of the 1:50 sectional model in Fig. 16(a) show that there is a small amplitude of the 0.080 m VIV at the 3° AoA, and there is a VIV with a small amplitude of 0.075 m at the 5° AoA; the other large-amplitude vibrations are eliminated. The test results of the 1:20 sectional model in Fig. 16(b) show that there is a small amplitude (0.050 m) vibration at 3° AoA, and the remaining vibrations exceeding the allowance limitation have completely disappeared. Therefore, the aerodynamic control measures based on the removal of the maintenance rails suppress the maximum vibration amplitude of the actual bridge below 0.070 m. Since the aerodynamic control measures of the flow suppressor are simultaneously added (see Fig. 13(b)), the vertical VIVs of the real bridge can be completely eliminated (see Fig. 17).

6. Additional structural damping

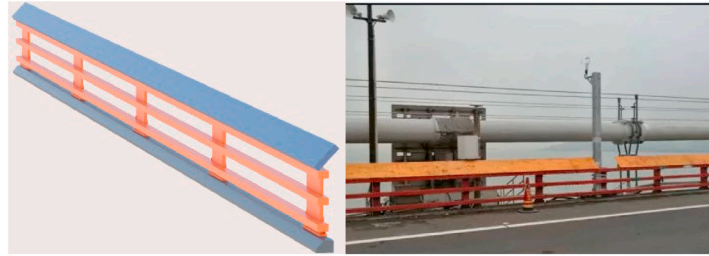
If the vortex shedding on the windward side of the bridge deck and bottom of the girder is the external cause of the VIVs (see Figs. 6 and 7), then the low structural damping would be the strengthening factor of the large-amplitude VIVs (see Table 2). Vortex shedding works through the internal low-damping condition, and is a reasonable explanation for the sudden first and succedent secondary VIV processes. If the damping ratio of the bridge structure is sufficiently large, VIVs should not occur. The Humen Bridge has not suffered from phenomena in the past 23 years of operation. To this end, an original cross-section of the real bridge and

Table 4
Comparison of VIV amplitudes and locked-in wind speeds between model and full-scale bridge.

Vertical mode	Angle of attack	Model amplitude/m	Mass correction/ η_m	Vibration mode correction/ η_φ	Actual bridge amplitude/m	Measured value/m	Corrected wind speed/(m/s)	Measured wind speed/(m/s)
The third-order	-3°	0.200	0.924	1.336	0.247	0.230	5.0–6.6	6.3–8.5
	0°	0.155			0.191		5.0–7.4	
	3°	0.245			0.302		6.6–8.3	
The fourth-order	-3°	0.200	0.949	1.280	0.243	0.278	6.0–8.0	8.3–9.6
	0°	0.155			0.188		6.0–9.0	
	3°	0.245			0.289		8.0–10.0	
The fifth-order	-3°	0.200	0.924	1.330	0.246	0.310	8.0–10.6	10.7–12.8
	0°	0.155			0.190		8.0–11.9	
	3°	0.245			0.301		10.6–13.3	

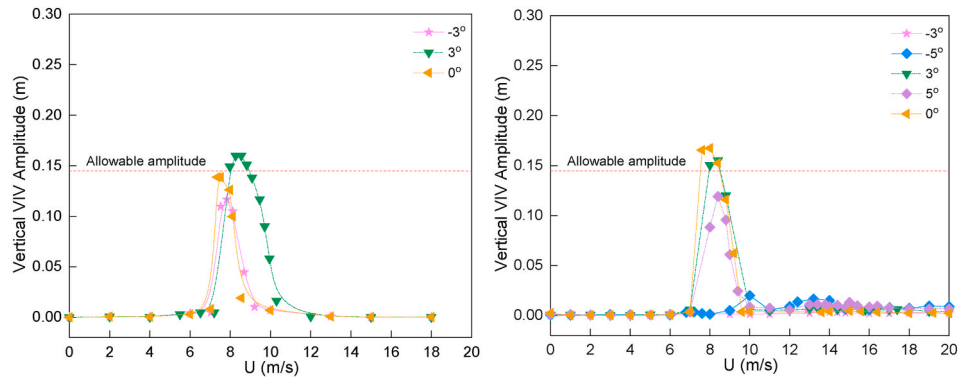


(a) Main girder and flow spoiler (unit: mm) (b) Local view of flow-suppressing plate (unit: mm)



(b) Rendering and on-the-spot installation of suppressing plate

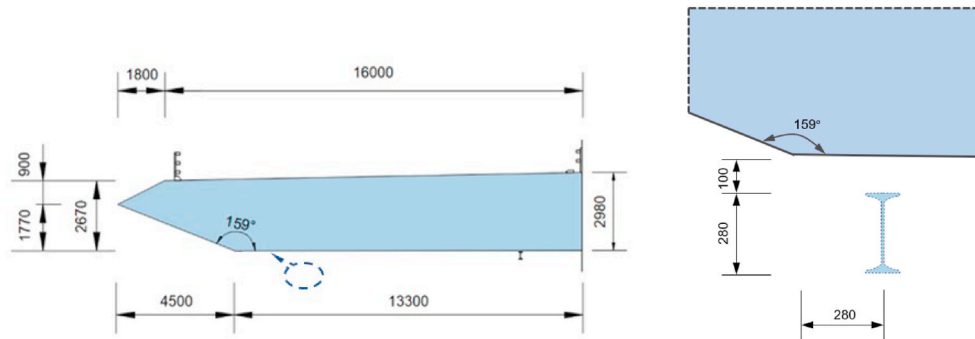
Fig. 13. Aerodynamic control measures of the flow suppressors.



(a) VIV of 1:50 model test

(b) VIV of 1:20 model test

Fig. 14. Wind tunnel test results of the VIV control effect of the suppressor plates.



(a) Main girder without outer maintenance rails

(b) Local view of maintenance rail

Fig. 15. Aerodynamic control measures of removal of outer maintenance rails (unit: mm).

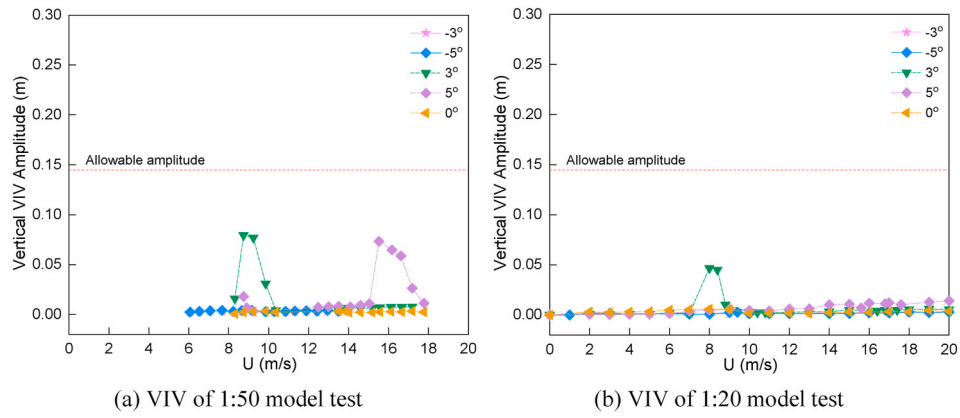


Fig. 16. Wind tunnel test results of VIV control effect of removal of maintenance rails.

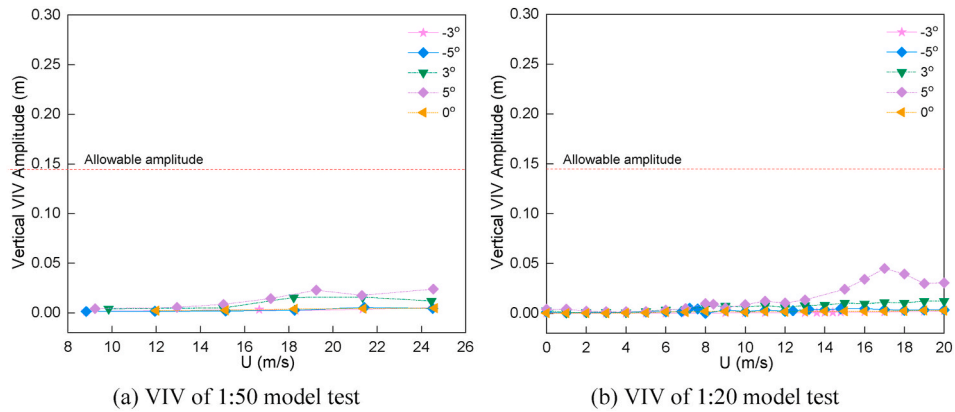


Fig. 17. Wind tunnel test results from adding flow suppressors and removing maintenance rails.

that with the addition of flow-suppressing plates were selected as case studies. The damping ratio of the structure was increased in the wind tunnel test of the sectional model, so as to investigate the VIV's damping-related sensitivity.

The damping control effect on the original cross-section of the real bridge was tested. The damping ratios of the four different cases were compared using the 3° AoA with the maximum VIV amplitude of the 1:50 sectional model, namely, the minimum damping ratio of the wind tunnel test was 0.32%, the minimum damping ratio increment was 0.43%, the recommended value of certain wind loading specifications was 0.50%, and the construction completion acceptance value was 0.63%. Fig. 18(a) depicts the VIV amplitudes at different damping ratios

at the 3° AoA. The amplitude at a damping ratio of 0.32% is 0.165 m. When the damping ratio increases to 0.43%, the amplitude is 0.055 m. When the damping ratio is further increased to 0.50%, the amplitude reduces to 0.025 m, and when the damping ratio increases to 0.63%, the VIV completely disappears.

As mentioned above, the effect of damping control after the addition of the flow suppressors was also tested. The 1:20 model with the 0° AoA was used to compare the damping ratios of three different situations: a minimum damping ratio of the wind tunnel test of 0.15%, 0.25%, and the recommended value of 0.50% from Chinese loading code. Fig. 18(b) depicts the VIV amplitudes at different damping ratios at the 0° AoA. The amplitude is 0.167 m when the damping ratio is 0.15%. When the

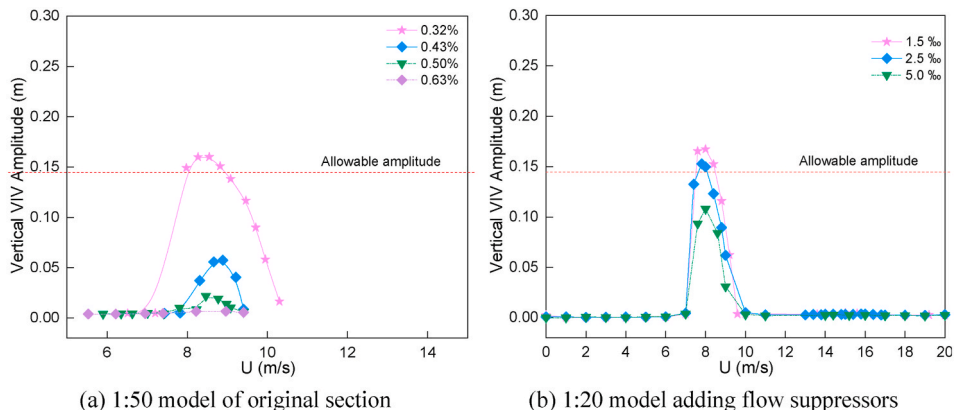


Fig. 18. Wind tunnel test results of damping control effects.

damping ratio increases to 0.25%, the amplitude is 0.153 m; when the damping ratio further increases to 0.50%, the amplitude becomes 0.108 m. The damping control effect is not as good as that of the 3° AoA of the 1:50 sectional model with the original cross-section. The actual bridge damping ratio is calculated as 0.25%, based on a mass correction coefficient of 0.924 and mode correction coefficient of 1.336. The actual bridge VIV amplitude is estimated as $0.153 \times 0.924 \times 1.336 = 0.189$ m, consistent with the observed results. The above tests showed that the secondary VIV could be related with the decreasing actual structural damping. The first large-amplitude VIVs of the Humen Bridge on May, 5 can lead to the reduction of the structural damping, deteriorating of potential VIV risk.

7. Conclusions

Although the VIVs of a suspension bridge will not cause safety problems such as structural damage or even collapse, they will cause bending or torsional single-mode vibrations of the main girder, and worsen the comfort of drivers or pedestrians on the bridge during the VIV process. In response to the sudden first and secondary VIV phenomena involving the high-order vertical modes of the Humen Bridge, the VIV triggering mechanisms were studied based on on-site acceleration measurements of the bridge girder and CFD numerical simulations. Among them, the sudden first VIV was caused by temporary water-filled barriers that changed the aerodynamic configuration of the box girder. After the water-filled barriers were removed, the VIV was effectively suppressed. The secondary VIV was caused by a reduction in the structural damping. Large-amplitude VIVs can lead to the reduction of the structural damping, deteriorating of potential VIV risk. Increasing the structural damping could reduce the VIV amplitude, and even suppress the VIV response.

A sectional model wind tunnel test was used to develop aerodynamic control and damping control measures. The aerodynamic control measures included the addition of flow suppressors implemented in the early stage and the removal of maintenance rails implemented in the later stage. These measures were successfully applied to the Humen Bridge; they effectively suppressed the VIV resonance, and quickly restored the bridge to traffic. The developed aerodynamic control measures and damping control measures will ensure that the Humen Bridge no longer suffers from large-amplitude VIVs, while improving the comfort of drivers and pedestrians.

CRedit authorship contribution statement

Yaojun Ge: Conceptualization, Formal analysis, Funding acquisition. **Lin Zhao:** Methodology, Calculation, Data curation, Writing – original draft, Writing – review & editing. **Jinxin Cao:** Test, Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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